

Enabling Under-Ice Geochemical Observations with a Size, Weight, and Power-Constrained Robot

Jess Horowitz, Department of Mechanical Engineering, Massachusetts Institute of Technology

Abstract: Whether the Arctic serves as a source or sink of emissions has significant implications on modeling climate change. Remotely operated vehicles (ROVs) can monitor dissolved methane (CH₄) and carbon dioxide (CO₂) below ice during the spring freshet. We demonstrate a size, weight, and power-constrained (SWAP) underwater vehicle (UV) and geochemical sampling system suitable for continuous under-ice geochemical observations in field-analog settings.

Keywords: *in situ sampling, geochemical, ROV, under ice, mechanical design*

Motivation & Objectives

- Climate change accelerated by disrupted natural **source-and-sink cycles**¹
- Spring freshet drives over **90% GHG emissions** in some environments²
- Previous studies restricted to surface vehicles in **ice-free waters**²

We propose to use a **SWAP-UV** to enable under-ice deployment, and design a novel topside pumping mechanism (ITMSS) in order to utilize geochemical instrumentation.

System Design Overview

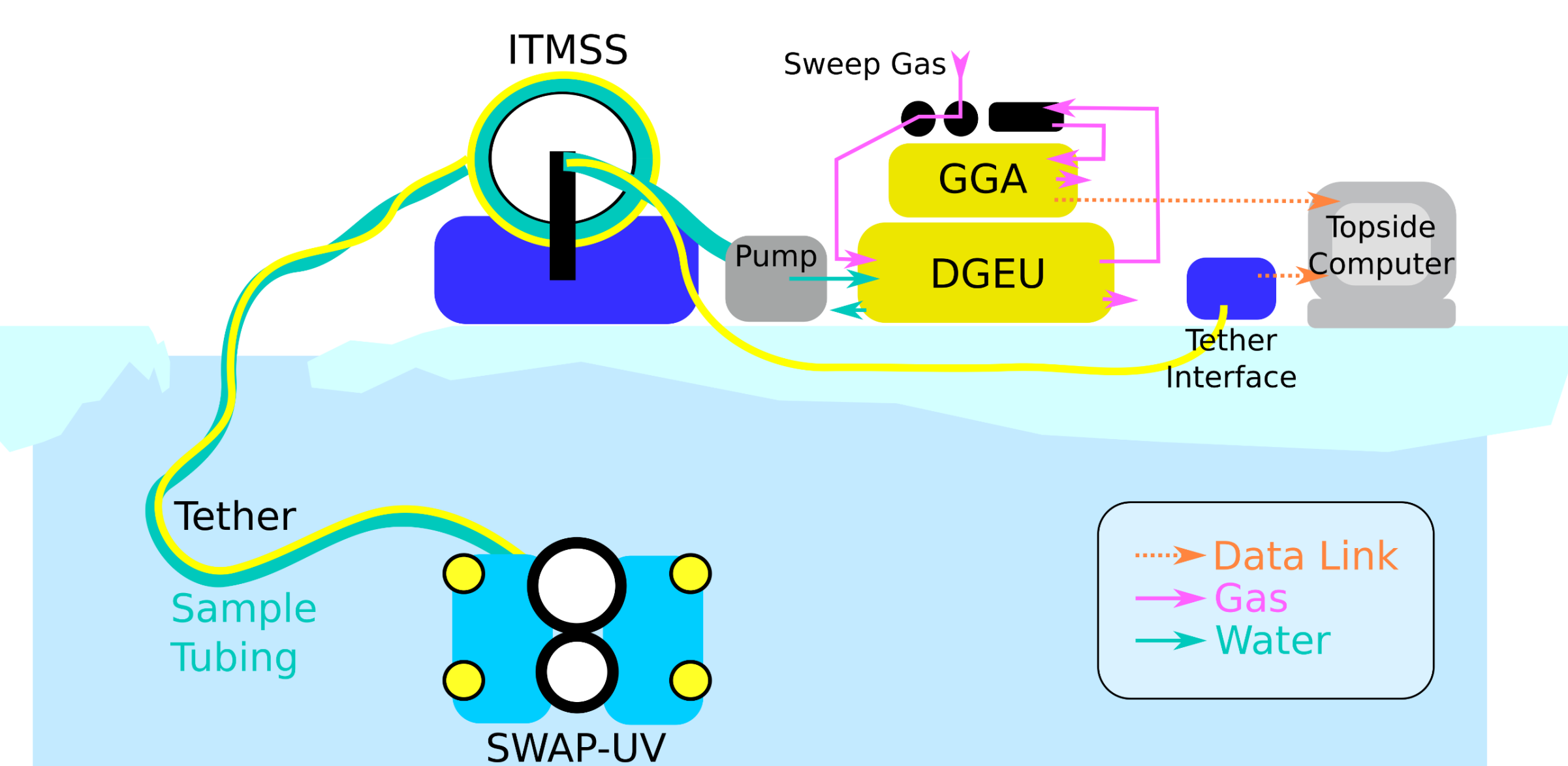


Figure 1: A conceptual illustration of the SWAP-UV, ITMSS, and topside geochemical laboratory and their connections during a field deployment for under-ice geochemical observation collection.

ITMSS Design Requirements

Qualitatively, the ITMSS must be modular, portable, allow for ease of maintenance, adaptable to various field operations, mountable to small vessel or ice-sled, mechanically sound, waterproof and corrosion resistant, and durable over many uses.

Attribute	Quantity
Minimum temperature	-15 °C
Maximum depth of SWAP-UV	5 m
Maximum horizontal radius	150 m
Maximum winch speed	1 m/s
Duration of winch operation	3 hours

References

- [1] V. Masson-Delmotte, et al. "IPCC, 2021: Climate Change 2021: The Physical Science Basis. Contribution of Working Group 1 to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change," 2021.
- [2] C. C. Manning, et al. "River inflow dominates methane emissions in an arctic coastal system," Geophysical Research Letters, vol. 47, no. 10, p. e2020GL087669, 2020

Integrated Tether Management Sampling System (ITMSS)

The ITMSS manages the long SWAP-UV tether and water collection tubing. The calculations for system specifications and the mechanical and electrical design are depicted below.

Component	Spec
DC Motor	0.5 hp, 1800 rpm
Gearbox	10:1
SLA Battery	12 V, 110 Ah
Diaphragm Pump	12 V, 5 A, 2 GPM

Motor Selection

$$F_D = \frac{1}{2} C_D A \rho V^2$$

$$\tau_o = \tau_f + R F_D$$

$$\Omega_o = \frac{V}{r}$$

$$P_\gamma = 4\Omega_o\tau_o$$

Designing for Arctic Environment

- Belt drive selected for corrosion and temperature resistance
- Fitted water-shedding polyethylene tarp and light-weight corrugated plastic
- Anodized aluminum pulley arm
- Wireless humidity monitor

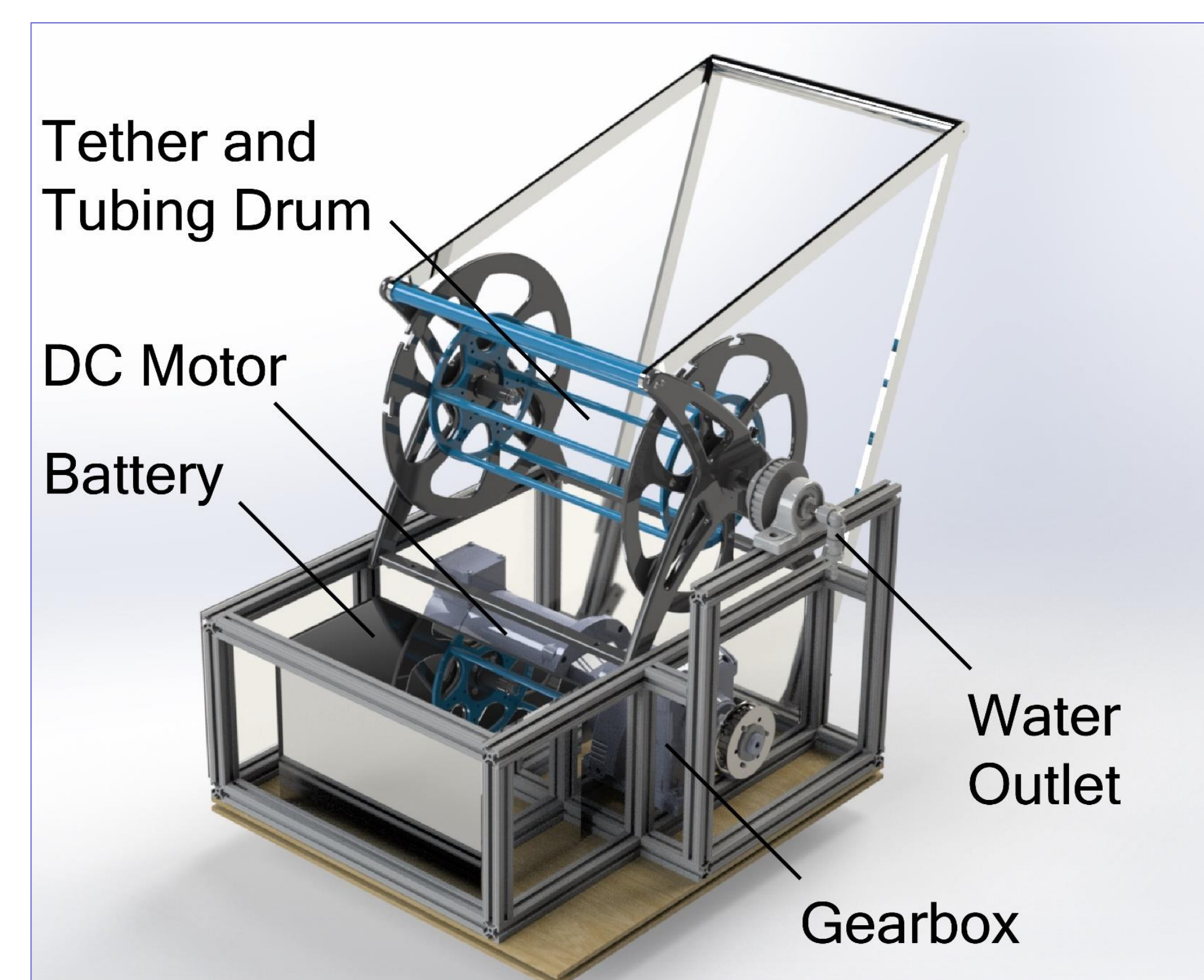


Figure 2: CAD rendering diagram of main TMSS components.



Figure 3: Photo of deployment and side view.



Figure 4: Cross section of ITMSS. Tethered electrical communications flow out of the drum through a slip ring. On the opposite side of the drum, water flows through linear low-density polyethylene (LLDPE) tubing within a hollow driven shaft.

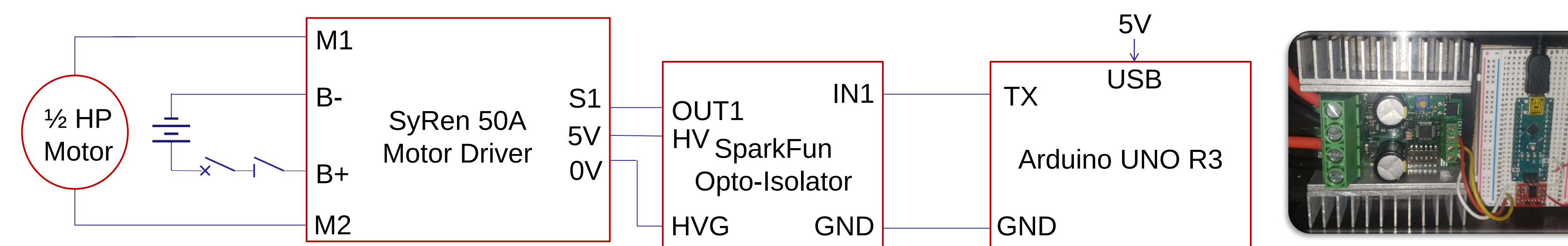


Figure 5: Electrical schematic diagram (left) and photograph of components inside waterproof junction box on ITMSS (right).

Arctic-Analog Fieldwork Results

We deployed at a frozen lake in MA to test the capability of measuring dissolved gases. Fig 6 and 7 show the results.

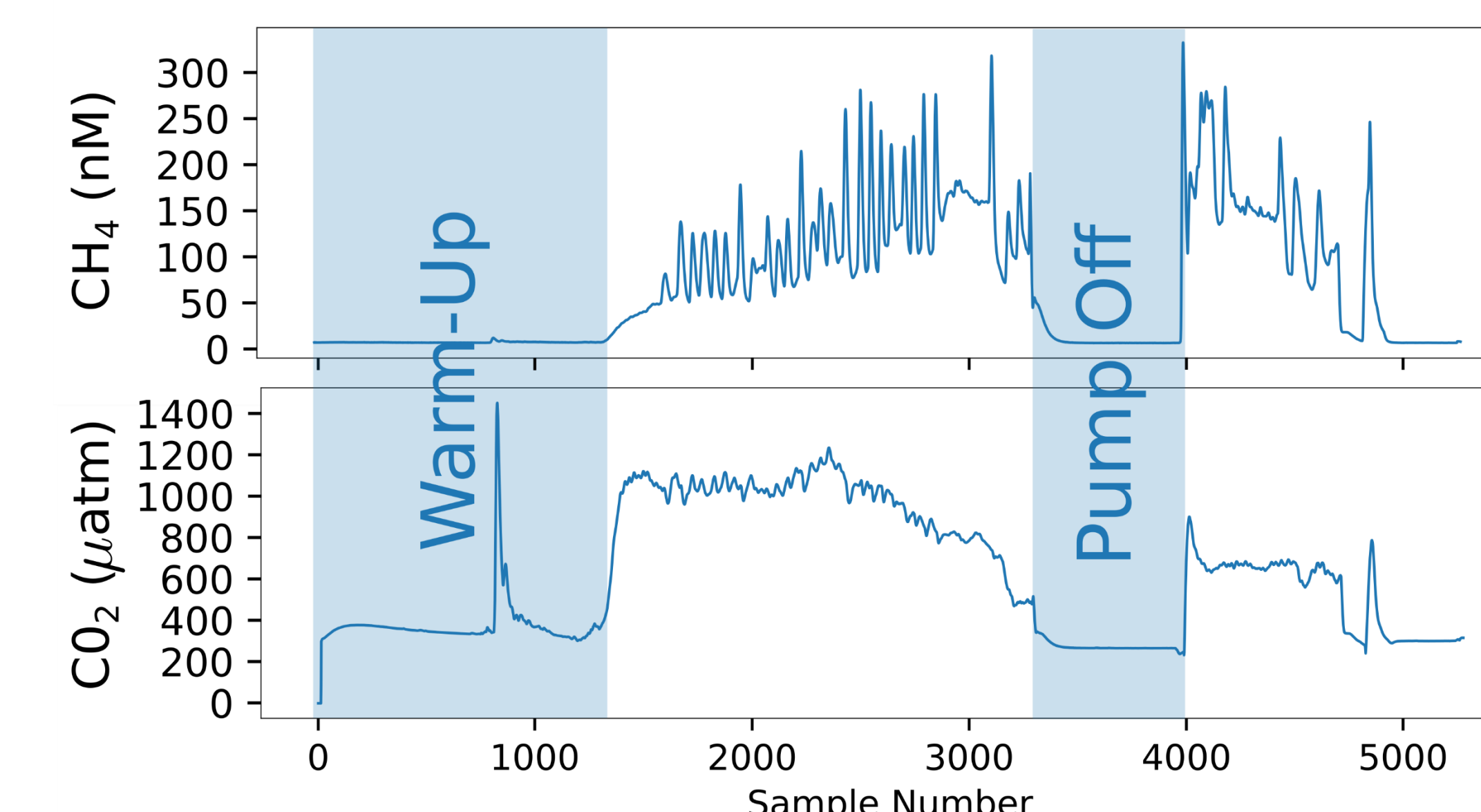


Figure 6: CH₄ and CO₂ measurements observed during the trials.

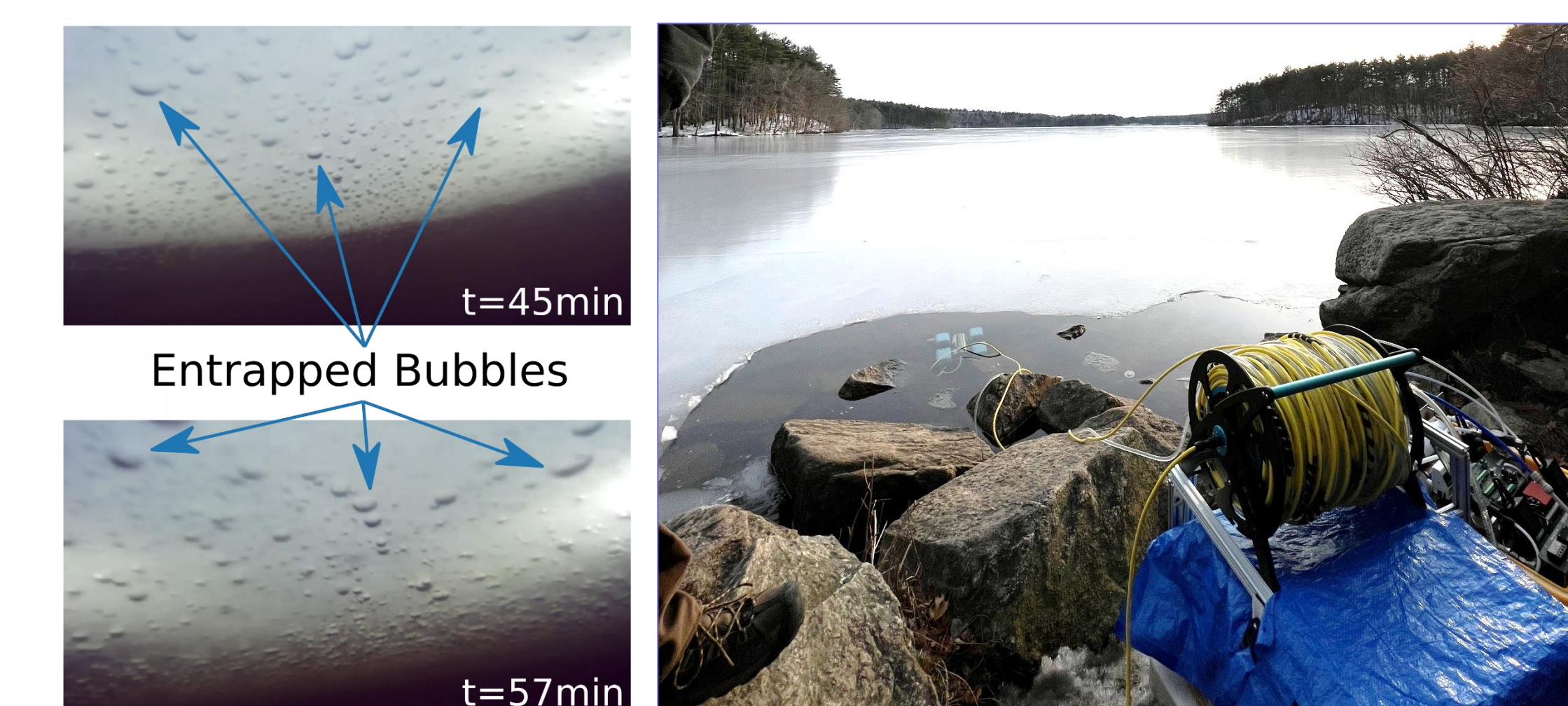


Figure 7: Image stills from recorded SWAP-UV video showing entrapped bubbles in the ice sheet (left). On-shore deployment point for under-ice work (right).

Conclusions

- We demonstrate a **functional modular system** for near real-time gas observations under ice with SWAP-UV
- The system measures interesting **gas structures** while filming qualitative **ice conditions**
- The system can be deployed in **Arctic environments**



Fig. 8 SWAP-UV in the Arctic, Summer 2022

Future work: Deploy the system in the Arctic and use data to examine the fate of methane during the spring freshet

Co-Authors and Affiliations

- Victoria Preston, MIT-WHOI Joint Program
- Anna P. M. Michel, Applied Ocean Physics and Engineering, Woods Hole Oceanographic Institution (WHOI)



Acknowledgements

Thank you B. Colson, S. Youngs, W. Pardis, and other members of the Michel lab at Woods Hole Oceanographic Institution for their assistance with field campaigns and system development. Thank you D. Fan and H. Schmidt at the MIT tow-tank for the use of their facilities for preliminary testing of the SWAP-UV. Thank you NSF OPP-1917528 for the grant for this project and ONR for their grant to the OCEANS program

